

9/29/26

T+P Week 2: Proof by Induction Alexander

1. Here is a counterexample to the proof:



Essentially, this is showing that the proof makes sense until you try to address the base case. We know the set of horses from H_1 to H_n where $n=1$ are the same color (one horse), and likewise for H_2 to H_{n+1} (just horse 2.) But when combining the two sets, no "middle" horses are the same color (there's no "middle" horses.) And, of course, $H_1 \neq H_2$ as well. Therefore, the proof is inconsistent.

2. Base Cases:

$$1=1 \quad 1+3+5+7=16$$

$$1+3=4 \quad 1+3+5+7+9=25$$

$$1+3+5=9 \quad \dots +11=36 \dots$$

It's pretty clear that $\sum_{i=1}^n 2i-1 = n^2$,
but we still need an inductive proof.
(cont.)

2. Cont.) Assuming the case for n is true, let's prove the $n+1$ case.

$$\sum_{i=1}^n 2i-1 + (2n+1) = (n+1)^2 = n^2 + 2n + 1.$$

Here, it's self-evident that $2n+1 = 2n+1$, and we know that $\sum_{i=1}^n 2i-1 = n^2$, the pattern works for $n+1$.

3. Base case: $1^3 = \frac{1^2(2)^2}{4} = 1^2 = 1. \checkmark$

Induction: Add the $(n+1)$ case for all 3:

$$1^3 + 2^3 \dots + n^3 + (n^3 + 3n^2 + 3n + 1) = \frac{(n+1)^2(n+2)^2}{4} = (1+2 \dots + 2n+1)^2.$$

Also... $1+2+3 \dots + n = \frac{n(n+1)}{2}$, so $1+2+3 \dots + n + (n+1) = \frac{(n+1)(n+2)}{2}$, and the 3rd expression equals $\frac{(n+1)^2(n+2)^2}{4}$. Second = is true.

$$\text{Next } \frac{n^2(n+1)^2}{4} + (n+1)^3 \text{ (next term in } \sum_{i=1}^n i^3 \text{ series)} \\ = \frac{n^2(n+1)^2}{4} + 4(n+1)^3 = \frac{(n+1)^2(n^2 + 4n + 4)}{4} = \frac{(n+1)^2(n+2)^2}{4},$$

exactly what happens when $n \rightarrow n+1$.

First = is true, too. Exp: $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4} = \left(\sum_{i=1}^n i\right)^2$.

4. Base case(s): $1^5 - 1 = 0 \checkmark$, $2^5 - 2 = 30 \checkmark$, $3^5 - 3 = 240 \checkmark$.

Inductive step: Say that $n^5 - n = 5m$.

Then, we have to show that $(n+1)^5 - n - 1 = 5(\text{something})$.

$$(n+1)^5 - n - 1 = n^5 + 5n^4 + 10n^3 + 10n^2 + 4n = n^5 + 4n + 5(n^4 + 2n^3 + 2n^2).$$

$n^5 + 4n = (n^5 - n) + 5n = 5(m+n)$, so also $5(\text{something})$.

Therefore, $(n+1)^5 - n - 1 = 5(\text{something})$, and Q.E.D.

*This is also why we need 2 base cases.

5. Base Case: $n=0$, $(1+x)^0 \geq 1+0x \checkmark (1=1)$.

Inductive Step: $(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) \dots$

$= nx^2 + nx + x + 1 \geq nx + x + 1 \quad (nx^2 \geq 0.)$ Thus,

$(1+x)^{n+1} \geq nx + x + 1 = 1 + (n+1)x$. Q.E.D.

6. By standard mathematical definition

(and convenience), let $\frac{1+\sqrt{5}}{2} = \phi$ (golden ratio)

and let $\frac{1-\sqrt{5}}{2} = \psi$ (golden ratio's "evil twin")

Base case: $n=0, n=1$

$n=0$: $\frac{1-1}{\sqrt{5}} = \frac{0}{\sqrt{5}} = 0$. $\checkmark F_n = 0!$

$n=1$: $\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \frac{2\sqrt{5}}{2} = \frac{2\sqrt{5}}{2\sqrt{5}} = 1$. $\checkmark F_n = 1!$ We have 2 bases now.

Inductive Step; strong-ish: we need $\frac{\phi^n - \psi^n}{\sqrt{5}}$ AND

$\frac{\phi^{n-1} - \psi^{n-1}}{\sqrt{5}}$ because $F_{n+1} = F_n + F_{n-1}$. Assume they're true.

Now, we must show that $F_{n+1} = \frac{\phi^{n+1} - \psi^{n+1}}{\sqrt{5}}$

$F_{n+1} = \frac{\phi^n - \psi^n}{\sqrt{5}} + \frac{\phi^{n-1} - \psi^{n-1}}{\sqrt{5}} = \frac{(\phi^n + \phi^{n-1}) - (\psi^n + \psi^{n-1})}{\sqrt{5}} = \frac{\phi^{n-1}(\phi+1) - \psi^{n-1}(\psi+1)}{\sqrt{5}}$

Recalling that both ϕ and ψ are roots

of the equation $x^2 - x - 1 = 0$, then $\phi^2 = \phi + 1$ and $\psi^2 = \psi + 1$.

Substituting this in: $\frac{\phi^{n-1}(\phi^2) - \psi^{n-1}(\psi^2)}{\sqrt{5}} = \frac{\phi^{n+1} - \psi^{n+1}}{\sqrt{5}}$

Now, we have shown that the equation

(Binet's Formula) holds by induction. Q.E.D.

7. Base case: $N=1$, $1 \cdot 1! = 1$. $\checkmark$

Inductive step: Assume $N = \sum_{i=1}^k a_i \cdot i!$. Prove $N+1$ works too.

Case 1: $N = a_k \cdot k! + \dots + a_1 \cdot 1!$, so $N+1$ adds 1 to

the sum. If $a_1 = 0$, it can become 1 to

fit the new number. Otherwise, $2!$ gets a

chance to take the carry-over from $1!$.

This keeps going until the 1 doesn't carry over... (cont.)

7. (cont), then the sum works for $N+1$.

Case 2: What if everything's full, i.e.

$k \cdot k! + (k-1)(k-1)! \dots + 1 \cdot 1!$. Here, when 1 is added, the whole sequence carries over, and it becomes $1 \cdot (k+1)! + 0 \cdot k! \dots + 0 \cdot 1!$.

This also works for $N+1$. Q.E.D.

Note: It turns out that $\sum_{i=0}^k i \cdot i! = (k+1)! - 1$.

Proof: $i \cdot i! = ((i+1) - 1) \cdot i! = (i+1)! - i!$.

$$\sum_{i=0}^k (i+1)! - i! = (2! - 1!) + (3! - 2!) + \dots + ((k+1)! - k!)$$

Every term cancels but $(k+1)!$ and $-1!$,

thus the sum equals $(k+1)! - 1$.

In the proof above, we can substitute Case 2's "filled system" with $(k+1)! - 1$, then adding 1 just gives $(k+1)!$.

8. Base case: $n=1$, number $\overline{aaa}$ sums digits to $3a$, so divisible by $3^1 = 3$. ✓

Inductive step: Number $\overline{bbbbbb \dots bbb}$ with 3^n digits (supposedly) is divisible by 3^n .

Now, with 3^{n+1} digits (essentially the number copied and shifted over twice),

we can factor it as (let's just call

our number b) $b \cdot \underbrace{100 \dots 00}_{3^n \text{ zeros}} \underbrace{100 \dots 00}_{3^n \text{ zeros}} 1$, where

the number of zeroes in each section are the digits of b , minus 1. The number

factored out is divisible by 3 ($1+0 \dots +0+1+0 \dots +0+1$),

and b is divisible by 3^n , so the new number is div. by $3^n \cdot 3 = 3^{n+1}$.

9. Base Case: $n=1, 5 \checkmark$ ($n=2, 25 \checkmark$ $n=3, 375 \checkmark$) ...
 Inductive step: Assume it works for 5^n .
 For 5^{n+1} , another odd digit must be placed to the left. In math, this can be expressed as $\text{new} = a \cdot 10^n + \text{old}$. The old number is already a multiple of 5^n , so we can factor this as $\text{new} = 5^n(a \cdot 2^n + k)$, where k is some positive integer. For new to be a multiple of 5^{n+1} , $(a \cdot 2^n + k)$ must be a multiple of 5. $0 = [1, 3, 5, 7, 9]$ has a remainder of $[1, 3, 0, 2, 4]$ when divided by 5 - every one possible - and multiplying by a power of 2 still gives all the remainders, because 2 and 5 are relatively prime. This guarantees an exact fit with whatever k is mod 5, thus new is divisible by 5^{n+1} . Q.E.D.

10. Base Case: $n=1, 1=F_1 \checkmark$ ($n=2, 2=F_2 \checkmark$ $n=3, 3=F_3 \checkmark$ $n=4, 1+3=F_1+F_3 \checkmark$)
 Inductive step: Suppose Zeckendorf's Theorem holds up to n . We must show that it works on $n+1$.
 Case 1: $n+1$ is a Fibonacci number
 Easy enough - just sum n to get $n+1$!
 Case 2: $n+1$ is not, so $F_k < n+1 < F_{k+1}$. Say that $r = (n+1) - F_k$. It must have a Zeckendorf solution, so $n+1 = (\text{zeckendorf} \checkmark) + F_k$. Now we must show that F_k is not consecutive with any F_m in the Zeckendorf solution for r . (cont.)

* In this case, it's the Zeckendorf solution.

10. (cont.) If it was, it would be with the largest F_m in r . $r = n+1 - F_k < F_{k+1} - F_k$, but by the definition of Fibonacci numbers, $F_{k+1} - F_k = F_{k-1}$. So, $r < F_{k-1}$. Therefore, r cannot have F_{k-1} in its representation, so F_k cannot be consecutive with an F_m in r . Thus, $n+1$ has a Zeckendorf solution of the form $F_k + r$. Q.E.D.

11. $m=0$: Trivial, always $n+1$ by definition.

$m=1$: $A(1,0) = A(0,1) = 2$. $A(1,1) = A(0, A(1,0)) \dots$

$= A(0,2) = 3$. $A(1,2) = A(0, A(1,1)) = A(0,3) = 4$.

The answers appear to be $n+2$. Proof?

Assume $A(1,n) = n+2$. Show that $A(1,n+1) = n+3$. ($= (n+1)+2$)

$A(1,n+1) = A(0, A(1,n)) = A(0, n+2) = n+3$. Q.E.D.

$m=2$: $A(2,0) = A(1,1) = 3$. $A(2,1) = A(1, A(2,0)) \dots$

$= A(1,3) = A(0, A(1,2)) = A(0,4) = 5$. The

pattern appears to be $2n+3$ for $m=2$.

PROOF by induction: Assume $A(2,n) = 2n+3$.

$A(2,n+1)$ must be $2(n+1)+3 = 2n+2+3 = 2n+5$, then.

$A(2,n+1) = A(1, A(2,n)) = A(1, 2n+3) = A(0, A(1, 2n+2))$.

$= A(1, 2n+2) + 1 \dots A(1,n) + n + 3 = (n+2) + n + 3 = 2n+5$. Q.E.D.

$m=3$: $A(3,0) = A(2,1) = A(1, A(2,0)) = A(1,3)$

$= A(0, A(1,2)) = A(0,4) = 5$. $A(3,1) = A(2, A(3,0))$

$= A(2,5) = 13$. $A(3,2) = A(2, A(3,1)) = A(2,13) = 29$.

Seems like $2^{n+3} - 3$. Prove for $2^{n+4} - 3 = A(3,n+1)$.

$= A(2, A(3,n)) = A(2, 2^{n+3} - 3) = 2(2^{n+3} - 3) + 3 = 2^{n+4} - 6 + 3 = 2^{n+4} - 3$. Q.E.D.

11. (cont.) $m=4$: $A(4,0) = A(3,1) = 13$. $A(4,1) = A(3, A(4,0)) = A(3, 13) = 2^{16} - 3 = 65533$. $A(4,n) = 2^{2^{2^{\dots^{2^2}}}} - 3$.
 $A(4,n+1) = 2^{2^{2^{\dots^{2^2}}}} - 3$. $A(4,n+1) = A(3, A(4,n)) = A(3, 2^{2^{2^{\dots^{2^2}}}} - 3)$.
 $= 2^{2^{2^{\dots^{2^2}} - 3 + 3}} - 3 = 2^{2^{2^{\dots^{2^2}}}} - 3$. QED (has $n+1$ 2's.)

$m=5$: $A(5,0) = A(4,1) = 65,533$. We can prove this via induction and substitution, but standard math notation doesn't work here (too big!).

12. Unfortunately, I ran out of time on this problem. I'd love to discuss it more later, possibly on Discord or next class.